

at eighty-five. Overall, taxes drop by 50 percent at retirement but then stay at this same inflation-adjusted level.

Basic Living Expenses: Basic living expenses—always adjusted for 3 percent annual inflation—are assumed to fall by 30 percent at retirement, then another 20 percent at age seventy-five, and another 10 percent at age eighty-five. Thus, by eighty-five, real spending on basic expenses has fallen to about 50.4 percent of its preretirement level ($0.7 \times 0.8 \times 0.9 = 0.504$).

Health Care: Health care has an inflation rate of 7 percent instead of 3 percent. Health-care expenses also increase with age, adjusting upward by 15 percent at sixty-five, 20 percent at seventy-five, and another 25 percent at eighty-five. By eighty-five, real health expenses are assumed to be 3.7 times larger than their preretirement value ($1.15 \times 1.2 \times 1.25 \times 1.07^{20} / 1.03^{20} = 3.7$), assuming a somewhat simplified overall inflation rate of 3 percent.

Leisure: Like health care, leisure carries an inflation rate of 7 percent. It increases by 50 percent at retirement as retirees set out to enjoy their new freedoms, then drops by 50 percent at seventy-five and another 75 percent at age eighty-five. By eighty-five, even with the higher inflation rate, leisure spending is only 40 percent of its preretirement level in terms of the overall price index ($1.5 \times 0.5 \times 0.25 \times 1.07^{20} / 1.03^{20} = 0.4$).

By capturing both the differential inflation rates and the changing dynamics by age, age banding provides a useful tool for planning long-term retirement budgets. With this approach, however, it is not obvious that retirement spending will decline with age. It may, but rapid growth in health-care expenses could lead to an overall increase in spending needs at the highest ages.

Four Types of Retirement Spenders

One final study should be considered to help shed light on retirement spending patterns and which default assumptions could be appropriate for different types of retirees. In August 2015, J.P. Morgan Asset Management released a study about retirement spending by Katherine Roy and Sharon Carson titled “Spending in Retirement.” Their data set provides a “big data” analysis of 613,000 US households led by people fifty-five or older who